MIL-B-8571D
16 October 1970
SUPERSEDING
MIL-B-8571C
30 March 1966

MILITARY SPECIFICATION

BAG, STORAGE, DRINKING WATER

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers plastic bags for the storage of drinking water.

1.2 Classification. Plastic bags shall be furnished in sizes having the following capacities, as specified (see 6.2):

Size A - 5-quart capacity
Size B - 3-pint capacity

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

L-P-375	Plastic Film, Flexible, Vinyl Chloride
UU-P-268	Paper, Kraft, Untreated, Wrapping
PPP-B-636	Box, Fiberboard
PPP-B-676	Boxes, Set-Up
PPP-T-45	Tape, Gummed, Paper, Reinforced And Plain, For Sealing And Securing

Military

MIL-W-530	Webbing, Textile, Cotton, General Purpose, Natural Or In Colors
MIL-T-5661	Tape And Webbing, Textile, Cotton Reinforcing, Woven
MIL-F-10884	Fasteners, Snap

STANDARDS

Federal

FED-STD-191	Textile Test Methods

Source: https://assist.dla.mil -- Downloaded: 202610-17T10:36Z
Check the source to verify that this is the current version before use.

Military

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Department of Health, Education, and Welfare; Consumer Protection and Environmental Health Service; Food and Drug Administration.

The Federal Food, Drug, and Cosmetic Act, and General Regulations for its Enforcement.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington DC 20402.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Association, ATTN: Tariff Order Section, 1616 P Street, N.W., Washington DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago IL 60606.)

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2) before production is commenced, five samples of the finished bag shall be submitted or made ready for the contracting officer or his authorized representative for inspection as specified in 4.4. The approval of the preproduction samples authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with the applicable provisions of this specification. The preproduction samples shall be manufactured by the supplier in the same facilities to be used for manufacture of the production items.

* 3.2 Design and construction. The design and construction of the bag shall be as shown on Figures 1 through 9. The seam construction shall be the flat type. All joints and sections shall be secured by means of heat-sealing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:36Z
Check the source to verify that this is the current version before use.

* 3.2.1 Mouth. The mouth of the plastic bag shall be reinforced with vinyl-resin strips 0.060 thick by 0.500 wide conforming to type II, class I of L-P-375 shown on Figures 1, 2, 6, and 7.

* 3.2.2 Bag. The bag shall be constructed of flexible vinyl film conforming to type II, class 1 of L-P-375 except that lacquer lifting and crocking shall not apply. The film thickness shall be as follows:

Bag material (sizes A and B)	0.020 ±.022 inch
Keeper material	0.040 ±.004 inch
Closing straps (sizes A and B)	0.040 ±.004 inch
Carring loop	0.060 ±.006 inch

* 3.2.2.1 Bondability. The bondability shall conform to L-P-375 except that the breaking strength shall be not less than 50 percent of the specified film strength to which it is heat sealed.

* 3.2.2.2 Film suitability. The materials used in the construction of the bag shall impart no objectionable taste or odor to the water contained therein. The vinyl chloride plastic film, antioxidants, and any other substances incorporated in the plastic shall be approved by the Food and Drug Administration for use in contact with drinking water (see 4.3.1).

* 3.2.3 Buckle. The buckle shall be fabricated from molded nylon and shall conform to the design and dimension shown on Figure 3.

* 3.2.4 Snap fasteners. The snap fasteners for the size B bag shall conform to style 2, size 1, construction A of MIL-F-10884, except that the button shall be finish 5 (brown) or finish 2 (black), the snap fasteners shall be finish 2 (black).

* 3.2.5 Strap, carrying. The carrying strap for the 3-pint waterbag shall be constructed of 3/4-inch tape conforming to type I of MIL-T-5661, except that only the following characteristics shall apply: Material identification, total warp ends, picks per inch, breaking strength, and weave. The ends of the carrying strap shall be dipped or sealed to prevent raveling. The finished carrying strap shall conform to Figure 4. As an alternate, the carrying strap of the 3 pint waterbag may be constructed of 3/4 inch tape conforming to type I, class 1 of MIL-W-530.

3.3 Performance.

3.3.1 Burst pressure. When tested as specified in 4.6.1, there shall be no evidence of seam separation or leakage at seams of the bag.

3.3.2 Drop test (size B). When tested as specified in 4.6.2, leakage of water, other than slight leakage through the lips, shall be cause for rejection.

3.3.3 Flexing. When flexed over a 1/8-inch mandrel under the conditions specified in 4.6.3, the bag and heat sealed areas shall show no evidence of cracking or splitting. The heat sealed areas tested should include the center of the body and the flexing of the lips in the opposite direction.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:36Z
Check the source to verify that this is the current version before use.

3.3.4 Blocking. When tested as specified in 4.6.4, the bag shall show no blocking greater than scale No. 1.

* 3.4 Identification of product. The bag shall be marked in accordance with MIL-STD-130. The height of the letters shall be a minimum of 3/16 inch. The nomenclature shall include the following specific information:

BAG, STORAGE, DRINKING WATER
Size (applicable size to be inserted)
Specification MIL-B-8571D
Contract No. _____

3.5 Workmanship. The bag shall be constructed in a thoroughly workmanlike manner and shall be free from any defects that might affect proper functioning in service.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Quality conformance inspections. Quality conformance inspections shall consist of examinations and tests listed in Tables I and II, which shall be performed on individual bags and lots to determine the conformance of the individual bag or lot with the requirements set forth herein. Unless otherwise specified, inspection shall be performed in accordance with MIL-STD-105.

* 4.2.1 Sample size. The sample size and acceptance criteria shall be as specified in Table I and Table II.

* 4.2.2 Performance test. The finished bag shall withstand the tests referenced in Table II.

TABLE I. Sample Size and Acceptance Criteria

Inspection or Characteristic	Examination Method	Inspection Level	AQL
Examination of bag	4.5.1 (Table IV)	II	2.5 (major) 6.5 (total)
Examination for dimension	4.5.2	S-2	2.5
Examination for count	4.5.3	S-2	2.5
Examination of preparation for delivery	4.5.4 (Table V)	S-2	2.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:36Z
Check the source to verify that this is the current version before use.

TABLE II. End Item and Material Tests

Characteristic	Rqmts Para	Test Method	Rqmt Appl To: Ind. Unit	Rqmt Appl To: Lot Avg	No of Def Per Unit	Results Rptd As: Pass or 1/Fail	Results Rptd As: Num to Nearest	Insp Level	AQL
Burst Pressure	3.3.1	4.6.1	X		1	X		S-1	2/
Drop Test (Size B only)	3.3.2	4.6.2	X		1	X		S-1	2/
Flexing	3.3.3	4.6.3	X		1	X		S-1	6.5
Blocking	3.3.4	4.6.4	X		1	X		S-1	6.5

1/ If failure is indicated, report description of failure.

2/ Acceptance number zero. Rejection number one.

* 4.3 Contractor's certification. Components and materials may be accepted on the basis of a contractor's certificate of compliance with Table III.

TABLE III. Components

Component	Requirement
Bag materials (see 3.2.2)	Physical requirements of 0.040 and 0.060 inch - thick vinyl
Buckle (see 3.2.3)	Composition
Buckle material (see 3.2.3)	Presence of fungi nutrients

* 4.3.1 FDA certification. The supplier shall submit proof of compliance that all materials and processes used in the manufacture of storage bags for drinking water comply with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (3.2.2).

* 4.4 Inspection for acceptance. Inspection shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

* 4.4.1 First article inspection. When required, inspection and testing of the first article sample shall be made of five completely fabricated bags for all provisions of this specification, applicable to end product examination and tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:36Z
Check the source to verify that this is the current version before use.

4.5 Examination.

* 4.5.1 Bag assembly. The bag assembly shall be visually examined for the defects listed in Table IV. Except for defects with an asterisk in the classification column, defects shall be classified as indicated. Defects with an asterisk in the classification column shall be classified as major defects when seriously affecting serviceability or appearance and as minor defects when affecting serviceability or appearance, but not seriously.

4.5.2 Examination for dimensions. The bag shall be examined for defects in dimensions. Any dimension that is not within the specified tolerance shall be classified as a defect.

4.5.3 Examination for count of bag in intermediate containers. Bags packaged for shipment shall be examined to determine conformance with package markings and specific quantity. The sample unit for this examination shall be one box. Any box containing less than the specified or marked quantity of bags shall be classified as a defect. The lot size shall be the number of intermediate containers.

4.5.4 Examination of preparation for delivery. An examination shall be made to determine if applicable packaging, packing, and marking requirements, specified in Section 5 of this specification have been met. Samples shall be examined to determine the existence of defects listed in Table V.

4.6 Test methods.

4.6.1 Burst pressure. The bag shall be filled with 5 quarts or 3 pints (as applicable) of water at 80° ±10° Fahrenheit (F) and then closed properly by folding the neck portion and securing the buckle assembly. The bag shall then be subjected to an external force of 75 ±2 pounds applied for 1 hour upon an area of 25 square inches of the surface of the bag. The bag shall be examined for failures and leakages. Slight leakage of water between the lips of the top closure shall not be considered as a failure of the bag under the conditions of this test.

4.6.2 Drop test. The 3-pint bag shall be filled with 3 pints of water at 80°F ±10°F and then properly closed by folding the neck portion and securing the buckle assembly. The bag shall then be dropped from a height of 32 feet to a smooth concrete surface. Failure of the seams or leakage of water shall be cause for rejection. A slight leakage through the lips is permissible.

* 4.6.3 Flexing. After exposure to a temperature of -40° ±3.6°F for 24 hours ±2 hours, the bag shall be flexed in two areas, i.e., in the center of the bag, 3/4 inch below the keeper assembly; and the center of the lips in the opposite direction. The bags shall be flexed 180 degrees over a 1/8 inch mandrel. Any evidence of cracking or splitting shall be cause for rejection.

* 4.6.4 Blocking. The bag shall be subjected to the blocking test specified in Method 5872 of FED-STD-191, except that the temperature shall be 130° ±1.8°F and the bag shall be folded once face-to-face with the bottom edge of the bag against the buckle of the bag. Blocking greater than scale No. 1 shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:36Z
Check the source to verify that this is the current version before use.

TABLE IV. Visual Examination

Examination	Defect	Major	See 4.5.1	Minor
Quality of plastic material (film and strips, buckle, strap, and belt loop)	Material not as specified	X		
	Tear, cut, hole, crack or break	X		
	Sharp crease, wrinkle, bubble, blister, or gouge		*	
	Greater than superficial cracks in buckle	X		
	Chipping, delamination, or abrasion		*	
	Crazed, pitted, or porous		*	
	Sheeting stained, dirty, or discolored		*	
	Plastic sheeting not translucent			X
Quality of cotton tape (applies to size B only)	Not specified material	X		
	Type not as specified			X
	Cut, hole, smash, tear or any other imperfection		*	
	Width or shade not as specified			X
	Ends of carrying strap not dipped or sealed to prevent raveling		*	
Quality of snap fasteners (applies to size B only)	Not specified type, size, or construction	X		
	Finish not as specified or area of no finish		*	
	Color not as specified			X
	Split, broken, bent out of shape, fractured, chipped, malformed, or otherwise impaired		*	
	Surface not smooth			X
	Evidence of rust or corrosion			X
	Fin, burr, sharp edges, or machine or die marks			X

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:36Z
Check the source to verify that this is the current version before use.

TABLE IV. Visual Examination (Cont'd)

Examination	Defect	Major	See 4.5.1	Minor
Design and construction	Not specified shape or contour		*	
	Varies from design or construction shown in specified drawing or approved sample		*	
	Not specified seam construction (flat or lap) or not the same as approved sample	X		
Workmanship and assembly	Open seam at perimeter of heat seal	X		
	Open seam other than at perimeter of heat seal		*	
	Width of heat seal around perimeter not as specified		*	
	Poor heat seal resulting in inadequate fusion		*	
	Seam puckered or distorted in heat sealing		*	
	Any patch, splice, or repair of plastic material	X		
	Any component missing or operation omitted	X		
	Component loose or not attached as specified		*	
	Component misplaced or not aligned, not serving intended purpose, e.g., fastening tape will not securely engage in buckle when bag is in closed position or snap fasteners will not engage to form carrying strap on size B	X		
	Snap fasteners mismated or reversed			X

Source: https://assist.dla.mil Downloaded: 2026-10-17T10:36Z
Check the source to verify that this is the current version before use.

TABLE IV. Visual Examination (Cont'd)

Examination	Defect	Major	See 4.5.1	Minor
	Snap fastener improperly set, not securely clinched, or fabric cut in clinching			X
	Male or female portion of snap fastener failing to effect a smooth and secure closure			X
	Male and female parts of snap fastener on strap cannot be closed or cannot be separated from closed position			
	Will not engage socket or otherwise defective	X		
	Snap fastener requires undue force to operate, but will function			X
	Any foreign matter or material on outside of bag that might cause damage to the material or affect intended use		*	
	Any foreign matter or material not readily removable on the inside of the bag that might cause damage to the material or have injurious effect to the user	X		
	The mouth of the bag has split, crack or fissure	X		
Identification marking	Omitted, incomplete, illegible, incorrect, not proper size, wrong color, or misplaced			X

Source: https://assist.dla.mil Downloaded: 2026-10-17T10:36Z
Check the source to verify that this is the current version before use.

TABLE V. Preparation for Delivery Defects

Examination	Defect
Marking (interior and exterior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application
Materials	Any component missing Any component damaged, affecting serviceability
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, loose strapping, inadequate stapling Bulging or distortion of containers
Number of bags and number of intermediate containers	Any box containing less than specified or marked quantity of bags Number of interior packages is more or less than required

* 4.6.5 Taste and odor test. A new bag shall be completely filled and emptied twice with distilled water. The bag shall be refilled for the third time with distilled water, and then properly closed by folding the neck portion and securing the buckle assembly. The bag shall be placed in an oven maintained at 100° ±2°F, for 24 ±2 hours. The bag shall then be removed, cooled to room temperature, and checked for development of objectionable odor or taste.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

* 5.1.1 Level A. Each bag with buckle and closing strap positioned flat shall be completely wrapped in Grade B, 30 pound minimum basis weight, kraft paper conforming to UU-P-268. Ten bags of one size only shall be placed flat in a snug fitting container conforming to type I, or II, variety 1, class A, style 3 or 4 of PPP-B-676. The bag shall be laid flat alternating each bag end to bottom. The boxes shall be closed with a minimum of 3 inch wide tape conforming to Type III of PPP-T-45.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:36Z
Check the source to verify that this is the current version before use.

5.1.2 Level C. The bag shall be packaged in accordance with the industry practice. The packaging shall be adequate to provide protection from deterioration and physical damage during direct shipment from supply source to the first receiving activity for immediate use. The method used shall assure shipping at the lowest practical commercial rates.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

* 5.2.1 Level A. Twenty boxes (200 bags) packaged as specified in 5.1, shall be packed in a box conforming to style RSC, grade V2s of PPP-B-636. The box shall be closed, waterproofed and reinforced in accordance with the requirements specified in the appendix of the container specification.

* 5.2.2 Level B. Twenty unit packages (200 bags) shall be packed in a shipping container conforming to style RSC, type CF or SF, class domestic, variety SW, grade 275 of PPP-B-636. Box closure shall be in accordance with the requirements of method II as specified in the appendix of the box specification.

* 5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the container specification.

* 5.2.3 Level C. The bags, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at lowest rate. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

* 5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129. The identification shall include the following information in the order listed:

BAG, STORAGE, DRINKING WATER
MIL-B-8571D
Size _____
Stock No. _____
Contractor _____
Contract No. _____

6. NOTES

* 6.1 Intended use. The bags are intended to be included in military personnel survival kits as storage containers for fresh water, for stowing miscellaneous items which must be kept from salt water, or they may be inflated, tied together, and used for flotation purposes, such as survival or swimming.

* 6.2 Order data. Procurement documents should specify:

a. Title, number and date of this specification.

b. Size and quantity or bags required (see 1.2).

c. First article, when preproduction sample is required (see 3.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:36Z
Check the source to verify that this is the current version before use.

d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

e. Certificate of compliance for safety of materials and processes used in the manufacture of the storage bags.

* 6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - GL
Navy - AS
Air Force - 82

Review Activities:
Army - GL, MD
Navy - AS
Air Force - 82

Preparing Activity:
Air Force - 82

Project No. 8465-0487

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:36Z
Check the source to verify that this is the current version before use.

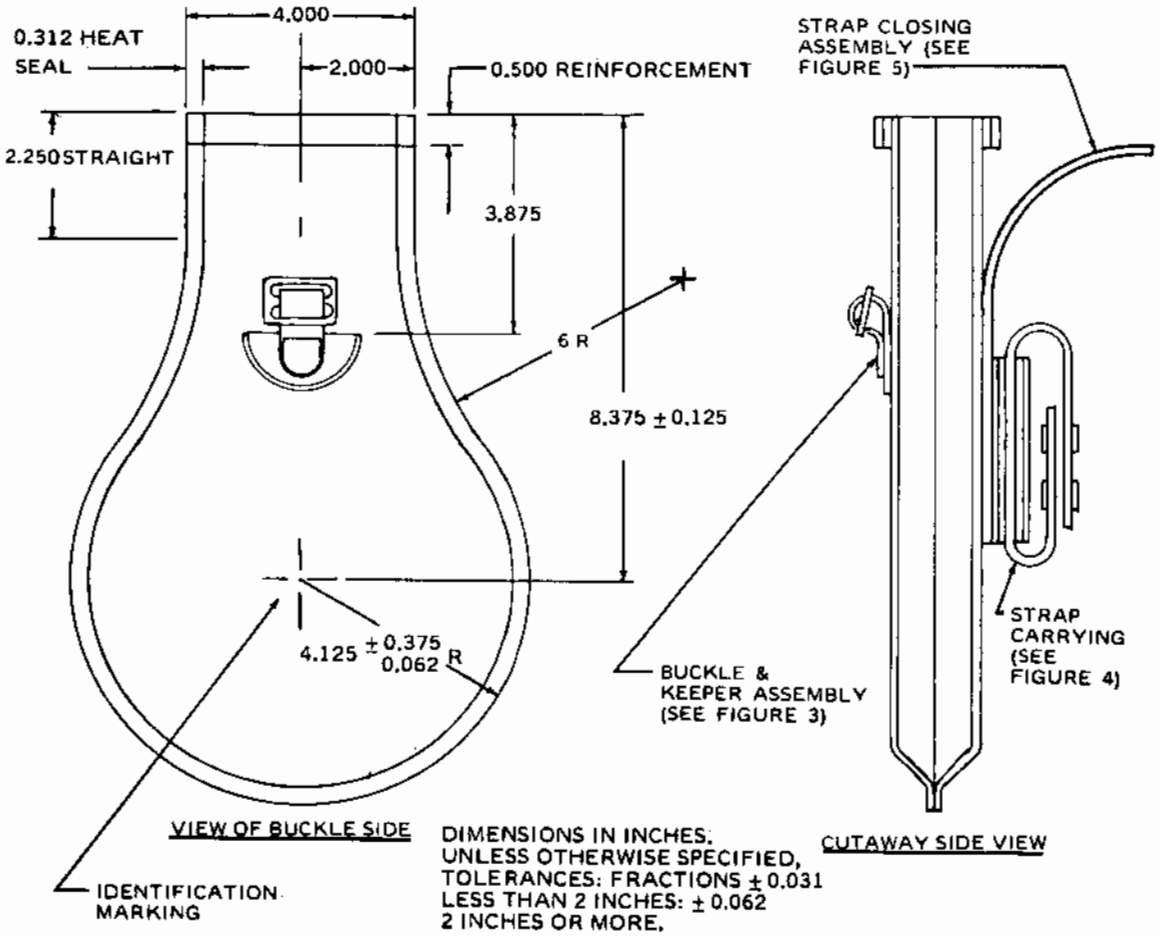


FIGURE 1. WATERBAG, SIZE B, 3 PINTS

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:36Z
Check the source to verify that this is the current version before use.

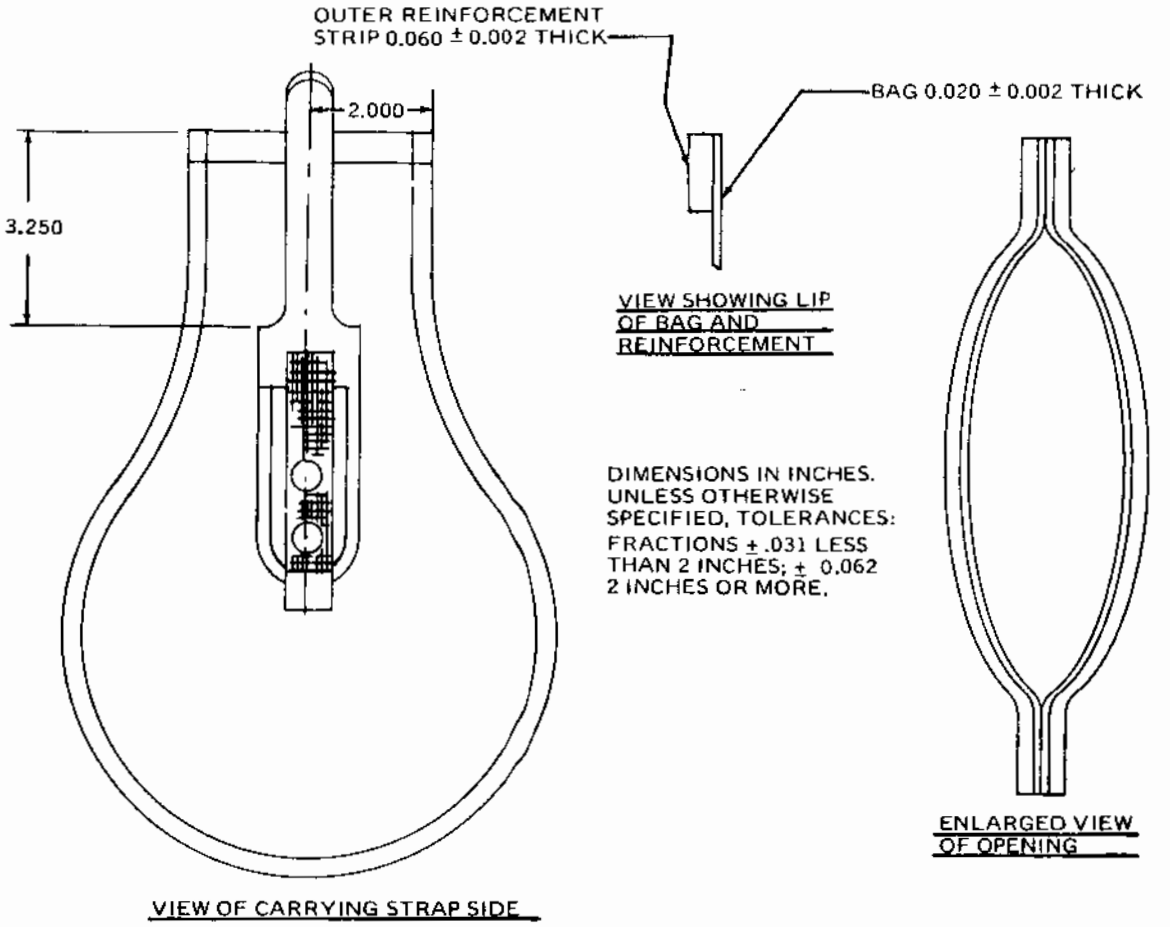


FIGURE 2. WATERBAG, SIZE B, 3 PINTS

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:36Z
Check the source to verify that this is the current version before use.

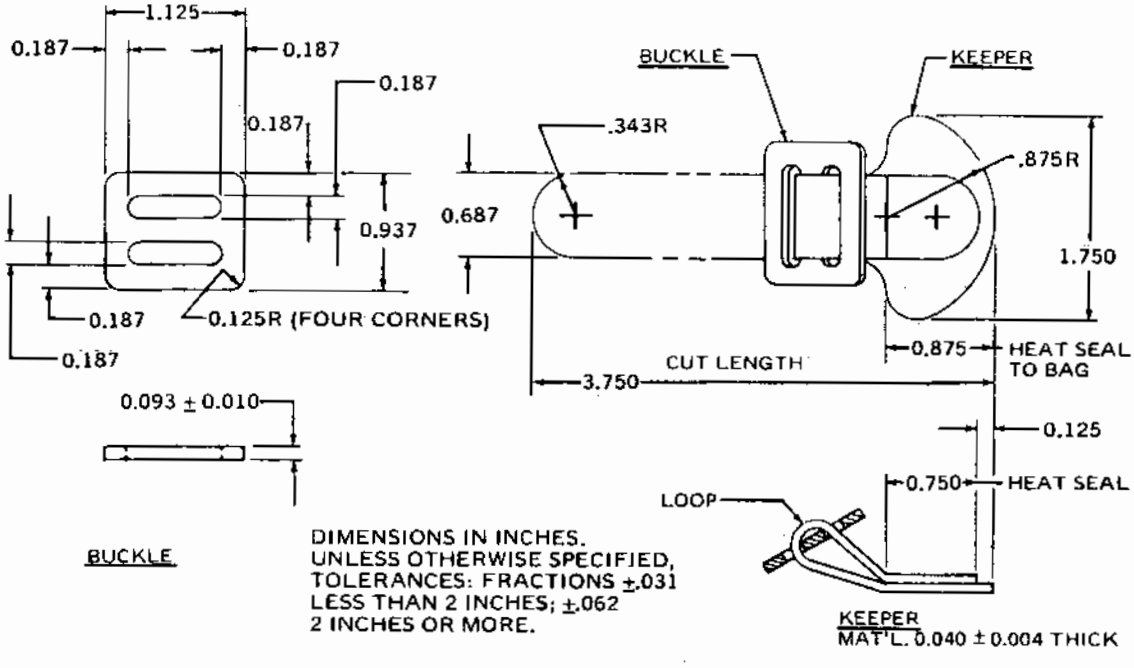
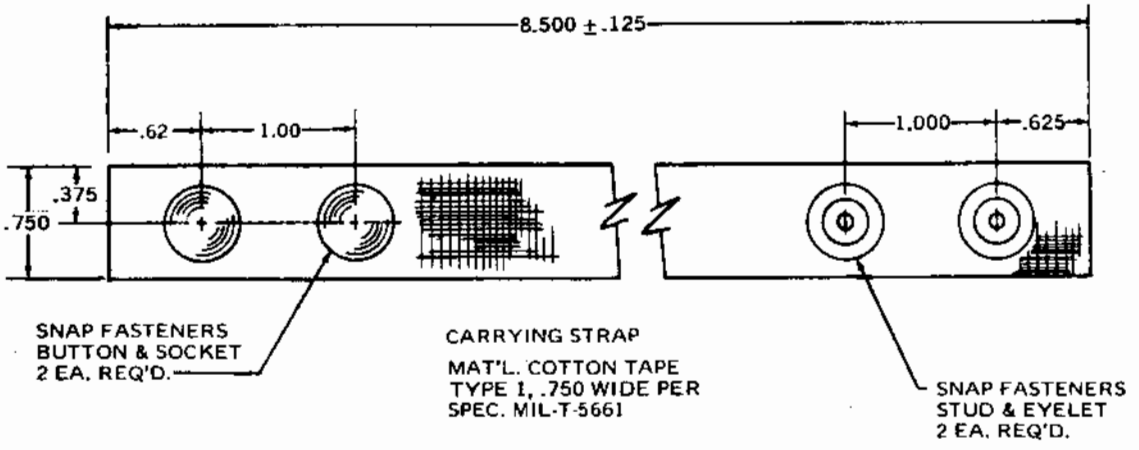


FIGURE 3. BUCKLE AND KEEPER ASSEMBLY BAG A, & B

Source: https://assist.dla.mil Downloaded: 2026-10-17T10:36Z
Check the source to verify that this is the current version before use.



DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERANCES: FRACTIONS ± .031
LESS THAN 2 INCHES; ± .062 2
INCHES OR MORE.

FIGURE 4. WATERBAG, SIZE B, 3 PINTS

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:36Z
Check the source to verify that this is the current version before use.

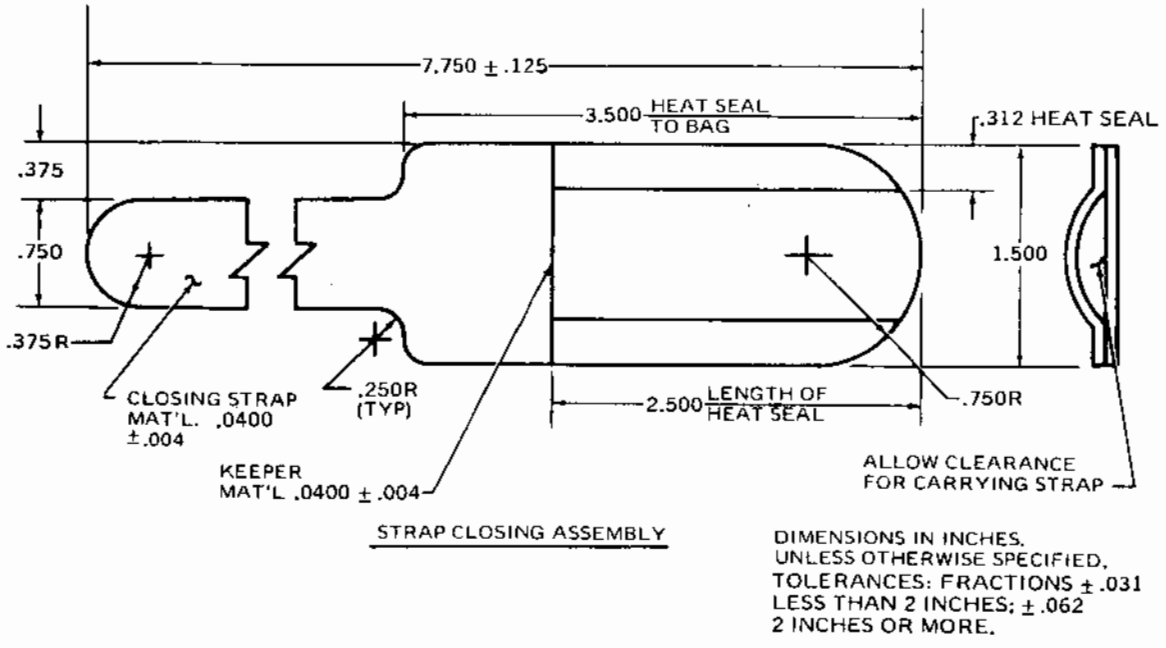
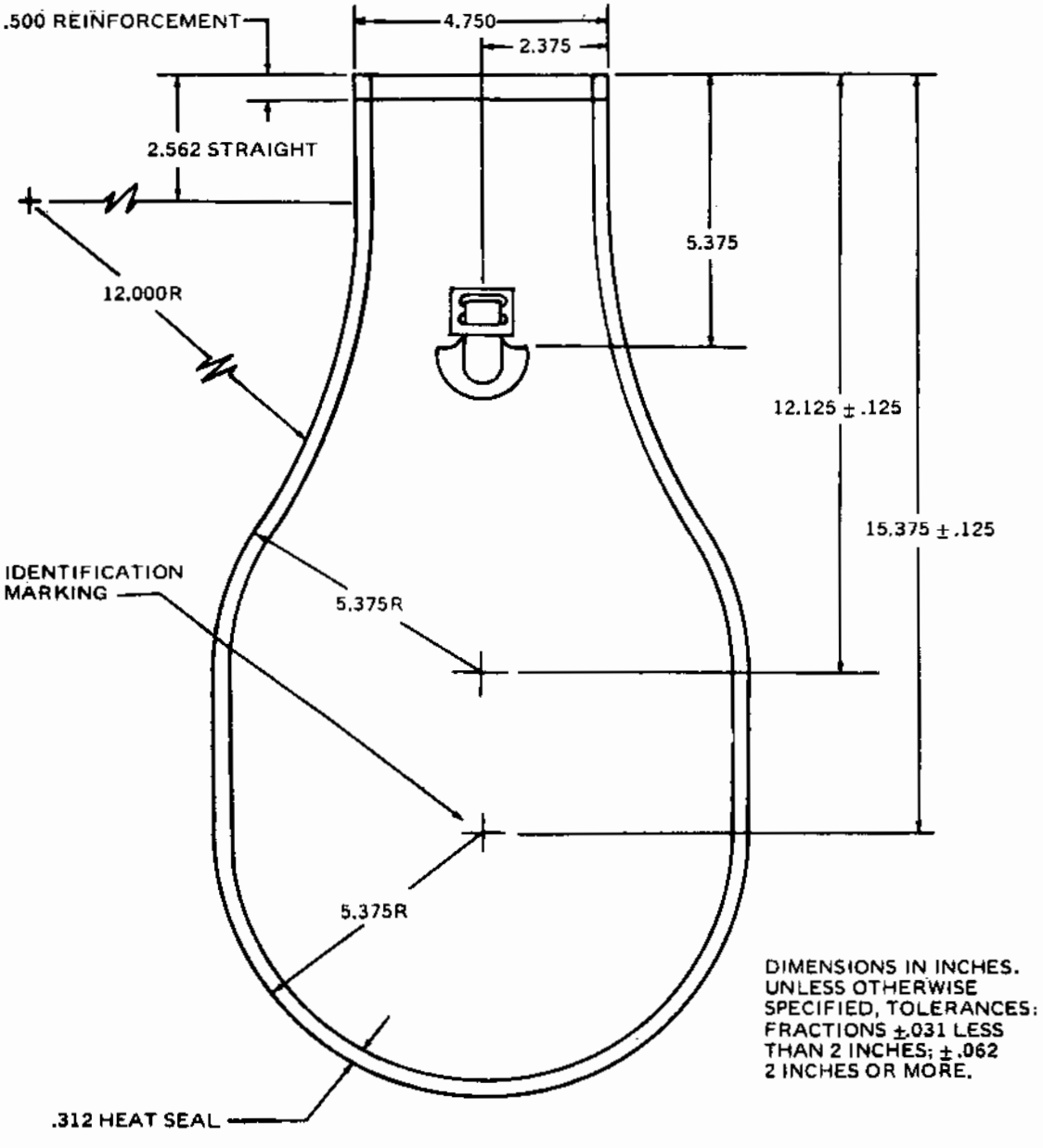


FIGURE 5. WATERBAG, SIZE B, 3 PINTS

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:36Z
Check the source to verify that this is the current version before use.



VIEW OF BUCKLE SIDE

FIGURE 6. WATERBAG, SIZE A, 5 QUARTS.

Source: https://assist.dla.mil -- Downloaded: 2026-01-17T10:36Z
Check the source to verify that this is the current version before use.

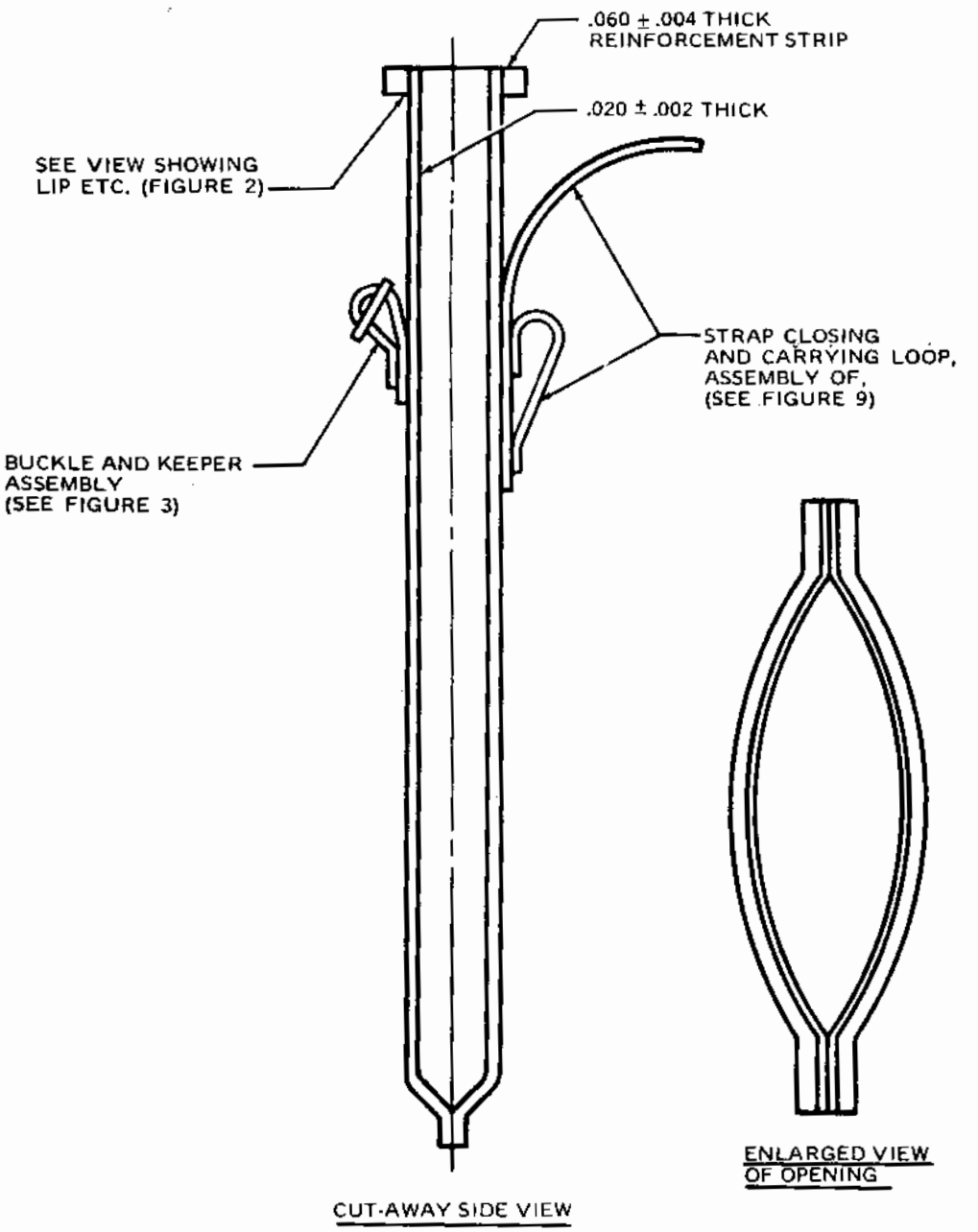
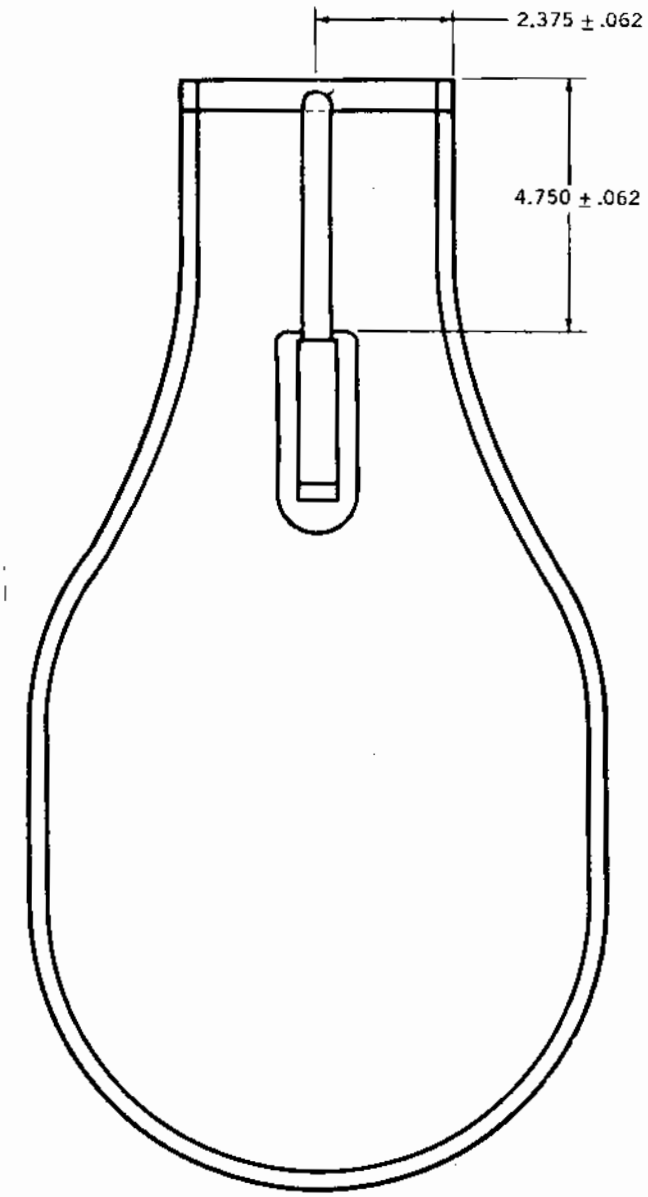


FIGURE 7. WATERBAG, SIZE A, 5 QUARTS.

Source: https://assist.dla.mil -- Downloaded: 2026-01-17T10:36Z
Check the source to verify that this is the current version before use.



VIEW OF CARRYING LOOP SIDE

FIGURE 8. WATERBAG, SIZE A, 5 QUARTS

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:36Z
Check the source to verify that this is the current version before use.

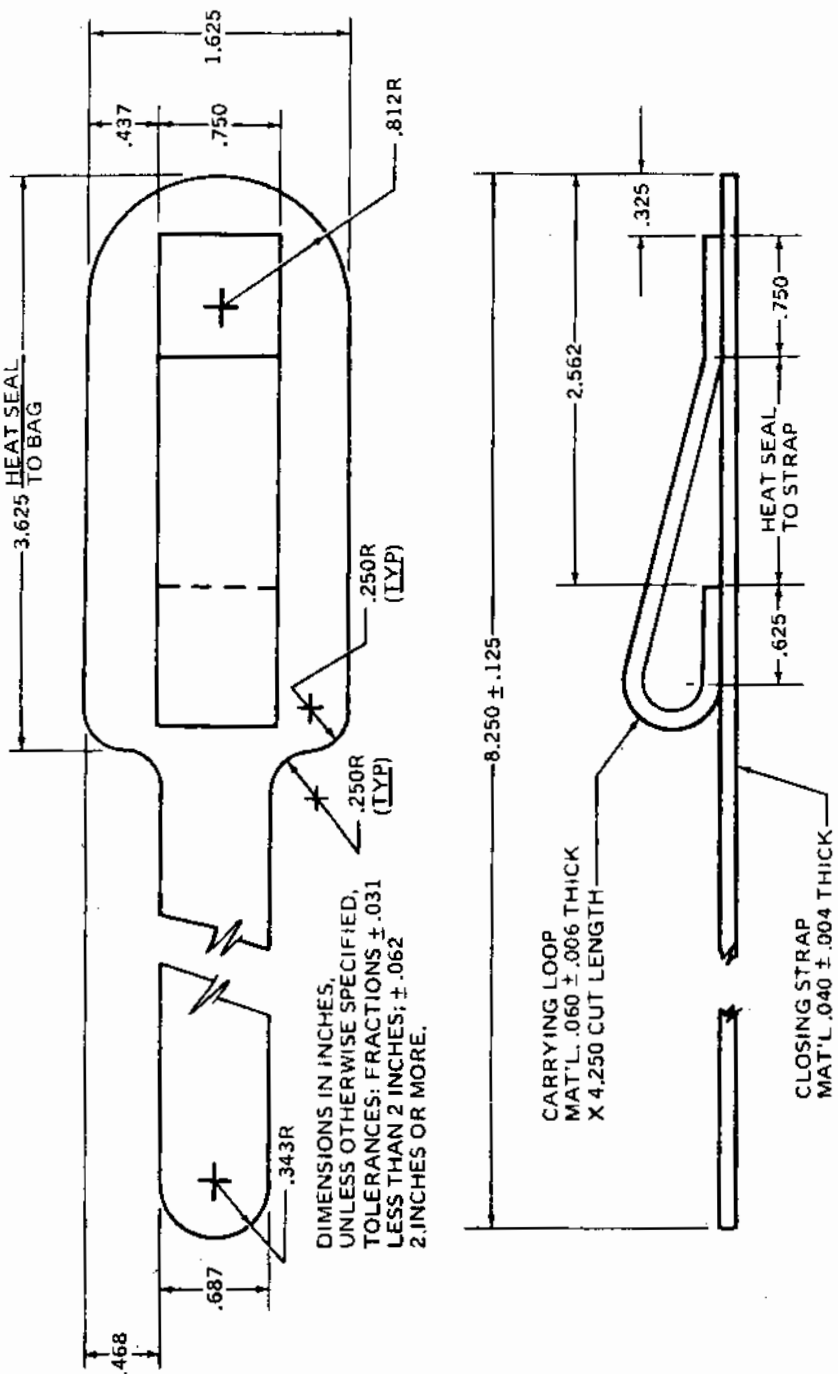


FIGURE 9. WATERBAG, SIZE A, 5 QUARTS

Source: https://assist.dla.mil/ -- Downloaded: 2026-10-17T10:36Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD, FORM 1 JAN 68 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:36Z
Check the source to verify that this is the current version before use.